Lista de Reglas de Deducción Natural

Reglas Básicas

Regla Trivial

$$\frac{\phi}{\phi}\ t$$

Conjunción

$$\frac{\phi \qquad \psi}{\phi \wedge \psi}\ i_\wedge \qquad \frac{\phi \wedge \psi}{\phi}\ e_{\wedge_1} \qquad \frac{\phi \wedge \psi}{\psi}\ e_{\wedge_2}$$

Disyunción

$$\frac{\phi}{\phi \vee \psi}\ i_{\vee_1} \qquad \frac{\psi}{\phi \vee \psi}\ i_{\vee_2} \qquad \frac{\phi \vee \psi \qquad \begin{array}{c}[\phi]^{(1)}\\ \vdots \\ \gamma\end{array} \qquad \begin{array}{c}[\psi]^{(2)}\\ \vdots \\ \gamma\end{array}}{\gamma}\ e_{\vee^{(1)(2)}}$$

Implicancia

$$\frac{\begin{array}{c}[\phi]^{(1)}\\ \vdots \\ \psi\end{array}}{\phi \to \psi}\ i_{\to^{(1)}} \qquad \frac{\phi \qquad \phi \to \psi}{\psi}\ e_{\to}$$

Negación

$$\frac{\neg\neg\phi}{\phi}\ e_{\neg\neg} \qquad \frac{\begin{array}{c}[\phi]^{(1)}\\ \vdots \\ \perp\end{array}}{\neg\phi}\ i_{\neg^{(1)}}$$

Bottom

$$\frac{\phi \qquad \neg\phi}{\perp}\ i_\perp \qquad \frac{\perp}{\phi}\ e_\perp$$

Reglas Derivadas

$$\frac{\phi \to \psi \qquad \neg\psi}{\neg\phi}\ MT \qquad \frac{\begin{array}{c}[\neg\phi]^{(1)}\\ \vdots \\ \perp\end{array}}{\phi}\ RAA^{(1)} \qquad \frac{}{\phi \vee \neg\phi}\ TND$$